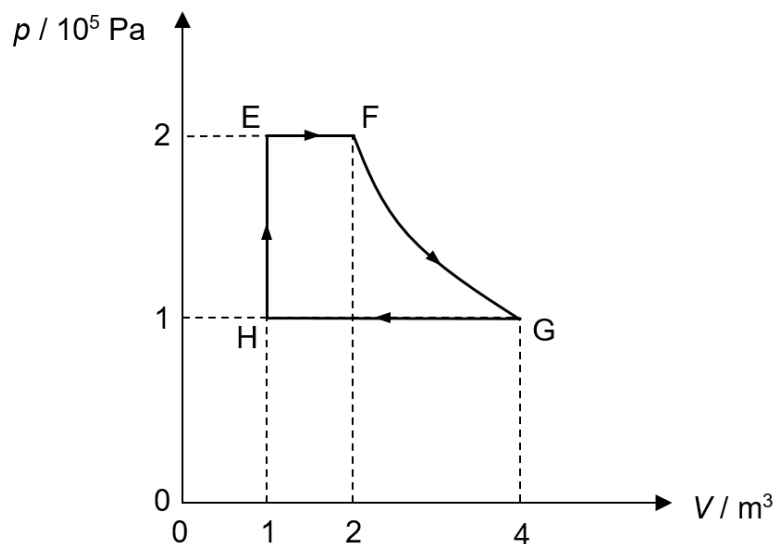


14 An ideal gas goes through the thermodynamic cycle EFGHE as shown.



Which of the following statements is correct?

- A After completing one cycle, there is net heat lost by the gas.
- B During process FG, there is no change in the internal energy of the gas.
- C During process HE, the gas loses heat.
- D The work done by the gas during process EF is equal in magnitude to the work done on the gas during process GH.

Ans: B

A is wrong. Net work done on gas is negative, in other words, there is a net work done by gas. As the change in internal energy is zero for one complete cycle, net work done by gas is equal to net heat gained by gas.

C is wrong. As work done on gas is zero, an increase in internal energy means that there is heat gained by gas.

D is wrong. Areas under the graph for EF and GH are different.

B is correct as FG is an isothermal process whereby temperature remains constant (since PV is unchanged)

15 This diagram shows the displacement-distance graph of a transverse wave at time $t = 0$. Taking